

CERTIFICADO DE CALIBRACIÓN

Cliente: Sociedad Minera Cerro Verde S.A.A.

RUC: 20170072465

Dirección: Cal. Jacinto Ibáñez Nro. 315 Urb. Parque Industrial, Arequipa.

Equipo: Detector Multigas

Marca: MSA

Modelo: ALTAIR 5XIR

Número de Serie: 108318

Sensores: LEL, O2, CO/H2S, CO2, NH3

Número de serie sensor: 00161202839, 01171055255, 04210855871, 10145738 08.21,
06210780316

Fabricado por: Mine Safety Appliances Co.

Estado Inicial: Operativo

Estado Final: Operativo

Fecha de calibración: 25/03/2025

Próxima calibración: 25/03/2026



Realizado: Dennis Huamanvilca
Supervisor Operativo



Revisado: Ing. Guillermo Cervetto.
Gerente General



I. Lugar de Calibración:

Sociedad Minera Cerro Verde.

II. Condiciones Ambientales:

Temperatura	19 °C
Presión	1025.1 hPa
Humedad Relativa	51 %

III. Patrones utilizados durante la calibración:

Patrón Utilizado	N/P	Lote N°	N° Certificado	Expiración
4G (CH ₄ , O ₂ , CO, H ₂ S) + SO ₂	10117738	403219836	1135622492	Dic-26
4G (CH ₄ , O ₂ , CO, H ₂ S) + CO ₂	10103262	402760900	1122489060	Ago-25
NH ₃	814866	403134822	1133335723	Mar - 26

IV. Resultados de Medición

Gas	Dosis del Gas	Equipos a Calibrar	% Error
		Lectura	
O ₂	15 %VOL	15 %VOL	0 %
CO	60 ppm	60 ppm	0 %
H ₂ S	20 ppm	20 ppm	0 %
LEL	58 %VOL	58 %VOL	0 %
CO ₂	2.5 %	2.5 %	0 %
NH ₃	25 ppm	25 ppm	0 %

**Realizado: Dennis Huamanvilca**
Supervisor Operativo**Revisado: Ing. Guillermo Cervetto.**
Gerente General

V. Anexos



CALGAZ,
A DIVISION OF AIRGAS USA LLC
321 Chesapeake Drive,
Cambridge, MD 21613
JSA Tel. 1-800-638-1197
www.calgaz.com

CERTIFICATE OF ANALYSIS

Date: 12/12/2024
Order Number: 1135622492
Lot Number: 403219836

Customer: MSA DEL PERU SAC
Part Number: 10117738
Use Before: DEC 2026

Component

Concentration
Accuracy: CH₄ ±5%, CO ±5%, O₂ ±5%, H₂S ±10%, SO₂ ±10%

METHANE
CARBON MONOXIDE
OXYGEN
HYDROGEN SULFIDE
SULFUR DIOXIDE
NITROGEN

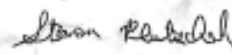
1.45%(29%LEL)
60PPM
15%
20PPM
10PPM
BALANCE

Cylinder Size: 2.0 Cu. Ft.
Contents: 58 Liter

Valve: 5/8"-18UNF
Pressure: 500 PSIG

Product composition verified by direct comparison to calibration standards traceable to N.I.S.T. weights and/or N.I.S.T. Gas Mixture reference materials.

Analyst:


Steve Plutschak


Scott Tingle



CALGAZ,
A DIVISION OF AIRGAS USA LLC
821 Chesapeake Drive,
Cambridge, MD 21613
USA Tel. 1-800-638-1197
www.calgaz.com

CERTIFICATE OF ANALYSIS

Date: 08/01/2023
Order Number: 1122489060
Lot Number: 402760900

Customer: MINE SAFETY APPLIANCE
Part Number: 10103262
Use Before: AUG 2025

Component	Concentration
	Accuracy: CH ₄ ±5%, O ₂ ±5%, H ₂ S ±10%, CO ±5%, CO ₂ ±10%
METHANE	1.45%(29%LEL)
OXYGEN	15%
HYDROGEN SULFIDE	20PPM
CARBON MONOXIDE	60PPM
CARBON DIOXIDE	2.5%
NITROGEN	BALANCE

Cylinder Size: 2.0 Cu. Ft.
Contents: 58 Liter

Valve: 5/8"-18UNF
Pressure: 500 PSIG

Product composition verified by direct comparison to calibration standards traceable to N.I.S.T. weights and/or N.I.S.T. Gas Mixture reference materials.

Analyst:


Steve Plutschak


Scott Tingle



CALGAZ,
A DIVISION OF AIRGAS USA LLC
821 Chesapeake Drive,
Cambridge, MD 21613
USA Tel. 1-800-638-1197
www.calgaz.com

CERTIFICATE OF ANALYSIS

Date: 09/17/2024
Order Number: 1133335723
Lot Number: 403134822

Customer: MSA DEL PERU SAC
Part Number: 814866
Use Before: MAR 2026

Component

Concentration
Accuracy: NH₃ ±10%

AMMONIA
NITROGEN

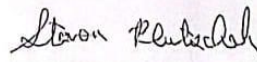
25PPM
BALANCE

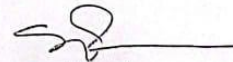
Cylinder Size: 2.0 Cu. Ft.
Contents: 58 Liter

Valve: 5/8"-18UNF
Pressure: 500 PSIG

Product composition verified by direct comparison to calibration standards traceable to N.I.S.T. weights and/or N.I.S.T. Gas Mixture reference materials.

Analyst:


Steve Plutschak


Scott Tingle